

Problem Chosen

2024

Team Control Number

A**HiMCM
Summary Sheet****15277**

This paper aims to assist the **International Olympic Committee (IOC)** in evaluating which sports, disciplines, or events (SDEs) should be added to or removed from the **2032 Summer Olympics** in Brisbane, Australia. We propose a comprehensive evaluation model to assess SDEs based on IOC's established criteria: popularity and accessibility, gender equity, sustainability, inclusivity, relevance and innovation, and safety and fair play. This evaluation aims to ensure that the Games remain relevant, impactful, and aligned with modern Olympic values.

To address the Olympic criteria, we identify various indicators and collect data of a diverse selection of Olympic sports. A multi-method approach is used combining the Entropy Weight Method (EWM) and Analytical Hierarchy Process (AHP) to assign weights to selected indicators. Since EWM and AHP are respectively objective and subjective methods, the results are then integrated using Grey Relational Analysis to create a comprehensive assessment of SDEs. The Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) is then employed to rank the SDEs based on their weighted overall performance across the criteria.

To test the applicability of the model, we collect objective data from official websites and subjective data from expert evaluations for a set of diverse SDEs including recently added or removed and long-standing sports. We compare our model's evaluation with the SDEs' Olympic legacy; as expected, long-standing sports are evaluated higher overall than new sports. The robustness of our model is also affirmed through a sensitivity analysis where the weights are adjusted.

With confirmation of the model's reliability, we identify potential candidates for inclusion in the **2032 Games** and evaluate them to find that **boxing, flying disc, and Australian rules football** have the highest ratings. Additionally, we use the **AutoRegressive Integrated Moving Average (ARIMA)** model to predict future SDE data and evaluate potential candidates for the **2036 Olympics**.

Finally, this paper includes a **non-technical letter** that communicates the model's rationale, results, and recommendations to the IOC.

Keywords: Olympic Games, Sports Evaluation, SDEs, Grey Relational Analysis, Entropy Weight Method, Analytical Hierarchy Process, TOPSIS, ARIMA

Contents

1	Introduction	4
1.1	Background	4
1.2	Problem Restatement	4
1.3	Problem Analysis	4
2	Assumptions and Justifications	5
3	Notations	6
4	Olympic Criteria Evaluation Model	6
4.1	Selection of Indicators	6
4.1.1	Popularity and Accessibility	7
4.1.2	Gender Equity	7
4.1.3	Sustainability	7
4.1.4	Inclusivity	7
4.1.5	Relevance and Innovation	7
4.1.6	Safety and Fair Play	7
4.1.7	Overall Indicator Framework	7
4.2	TOPSIS Analysis based on AHP-EWM Model	8
4.2.1	Analytical Hierarchy Process for Weight Calculation	9
4.2.2	Entropy Weight Method for Weight Calculation	12
4.2.3	Grey Relational Analysis for Weight Combination	13
4.2.4	TOPSIS Evaluation Model	14
5	Model Application and Results	16
5.1	Selection of SDEs	16
5.2	Evaluation of Olympic Status	17
5.2.1	Model Results	17
5.2.2	Evaluation Analysis	18
5.3	Sensitivity Analysis	19
6	Sports Evaluation for Future Olympics	20
6.1	Data Prediction with ARIMA	21
6.2	Evaluation Results	21
7	Evaluation of Strengths and Weaknesses	22

7.1	Strengths	22
7.2	Weaknesses	22
7.3	Future Improvements	23
8	Conclusion	23
9	A Letter to the IOC	24

1 Introduction

1.1 Background

Originated in Greece in 776 BC, the Olympics is an international sports event celebrated worldwide, showcasing athletic skills and promoting values such as unity, excellence, and fair play. **The International Olympic Committee (IOC)** continuously adapts the Olympic Games to reflect evolving global interests, ensuring that the Games remain relevant, inclusive, and impactful. Planning for the 2032 Summer Olympics in Brisbane, Australia, **the IOC aims to integrate sports, disciplines, and events (SDEs)** that resonate with contemporary values and attract a worldwide audience. **This paper considers numerous factors to create a mathematical model** that evaluates SDEs to provide recommendations for future Olympics.

1.2 Problem Restatement

Indicator Selection: Identify and describe the factors that need to be considered when assessing a sport's alignment with the IOC criteria, and justify the choices.

Model Building and Application: Using the factors identified above, formulate and analyze a mathematical model to further identify which SDEs align the most with the IOC criteria.

- a) Test the model on at least three SDEs that have been added or removed from the 2020, 2024, or 2028 Olympics and at least three SDEs that have been in the Olympics consistently since 1988 or earlier.
- b) Identify and order three SDEs that have the potential to be added or reintroduced to the 2032 Olympics by priority of consideration.
- c) Identify if any SDE has potential to be included in the 2036 Olympics.
- d) Evaluate the strength, weakness, and robustness of the model.

One-page Letter: Write a non-technical letter addressing the IOC with the above findings, including justifications to the model, recommendations for which SDEs to add or remove, and explanation for why the model supports these recommendations.

1.3 Problem Analysis

Indicator Selection: In addressing the IOC criteria for evaluating sports, we will identify major indicator categories and specify corresponding sub-criteria. Then, we need to

define how each sub-criteria can be quantified and data collection process.

Model Building and Application: We will develop an evaluation model to calculate the weights of each factor and determine the final ranking of SDEs. This model will assess new SDEs by calculating their scores and ranking them accordingly. Additionally, we will use a timeline-based forecasting model to predict the alignment of specific SDEs with IOC criteria for future Olympics based on historical trends and evolving factors.

One-page Letter: We will provide recommendations for SDEs and explain our mathematical modeling approach to optimize the selection of Olympic sports in an easily understandable language.

2 Assumptions and Justifications

Assumption 1. All data collected are accurate records of sports events.

Justification 1. The data is typically gathered from official sources, which are highly reliable. Any occasional minor inaccuracies that might occur are beyond our control and are unlikely to significantly impact the overall evaluation.

Assumption 2. Geopolitical factors, such as international conflicts or policy changes, will not significantly impact the inclusion of sports in the 2032 Olympics.

Justification 2. While geopolitical events may influence the Olympics, our study focuses on stable factors like sustainability and gender equity, which are enduring concerns and less subject to sudden change. Assuming no significant changes happen to the decision making of future Olympics allows our modeling to be feasible.

Assumption 3. The evaluation criteria for Olympic sports will align with data and metrics from the largest international leagues and competitions for each sport.

Justification 3. International leagues reflect a sport's global reach, popularity, and competition level, providing a reasonable basis for Olympic evaluation.

Assumption 4. The features of each sport for the 2032 Olympics will remain consistent with the 2024 data.

Justification 4. Olympic sports evolve gradually, so data in 2024 serves as a reliable approximation for 2032. Using a prediction model only for each Olympic event count, for which data over years is provided, reduces the complexity of data collection and evaluation.

Assumption 5. The relative importance of each feature will be consistent across all sports considered for inclusion in the 2032 Olympics.

Justification 5. This assumption ensures uniform application of criteria, simplifying the model and providing consistency in evaluation.

3 Notations

The definitions of used symbols in modeling are listed in Table 1.

Table 1: Symbols and Definitions

Symbol	Definition
$N_{\text{countries}}$	Number of participating countries
Y_{youth}	Proportion of young audience
GF	Government funding for the sport
T_{cycle}	Time since sport's introduction
G_i	Geometric mean of i
$I(p)$	Evaluation criteria for p
X_i	Raw data set
Y_{ij}	Normalized value
E_j	Information entropy
p_{ij}	Probability of ij
W_{final}	Final composite weight
$W(t)$	Weighted normalized decision matrix
W_j^+	ideal solution assigned to j -th criteria
W_j^-	anti-ideal solution assigned to j -th criteria
D_i^+	Distance from ideal solution to i -th alternative
D_i^-	Distance from anti-ideal solution to i -th alternative
C_i	Approximate degree of i -th alternative
ψ_0	Trend component
ϵ_t	Random error at time t
ϵ_{t-j}	Lagged value of error j at time t
$\hat{X}$	Predicted value
GRA_i	GRA value of i
W'_i	Normalized AHP weights for i
e'_j	Normalized EWM weights for j
ζ	Distinguishing coefficient
α_i	Weight coefficient of i

4 Olympic Criteria Evaluation Model

4.1 Selection of Indicators

In order to build a model that evaluates each SDE's alignment with Olympic criteria, we summarize the IOC's objective^[1] into six main indicators. The reasoning for each factor is explained in the section below.

4.1.1 Popularity and Accessibility

Popularity refers to the level of public interest and cultural relevance a sport holds on a global scale, while accessibility refers to how easily people can participate in or follow a sport. Together, they indicate how widely the sport is followed and played across different countries and cultures.

4.1.2 Gender Equity

The IOC values the fair treatment, access, and opportunity provided to athletes of all genders. Historically, female athletes have faced systemic barriers and limited access to competitive sports and equitable treatment.

4.1.3 Sustainability

Sustainability refers to the mindful use of resources to meet present needs without compromising their availability for future generations. The IOC aims for the minimization of waste and promotes awareness to green alternative energy.

4.1.4 Inclusivity

Inclusivity evaluates how well a sport represents diverse cultures and promotes widespread global participation. An inclusive sport not only is accessible to disadvantaged groups, but also welcomes participants and audience regardless of geographic barriers.

4.1.5 Relevance and Innovation

The use of technical tools, such as Hawk-Eye and instant replay systems, can enhance fairness and transparency of games. More extensive incorporation of latest technologies like artificial intelligence and e-sports integration can attract a larger audience, especially the newer generations.

4.1.6 Safety and Fair Play

Ensuring a safe and fair competitive environment is essential to maintain the credibility of Olympic sports. Strict anti-doping measures and judging systems minimize unfairness and errors. Lower safety risks ensure athlete well-being during competitions.

4.1.7 Overall Indicator Framework

A more detailed indicator framework is shown in Table 2 with a second layer of indicators and respective explanation.

Table 2: Indicator Framework Table

Level I	Level II	Explanation
Popularity and Accessibility	Global audience interest	Google interest trend
	Athlete participation	Number of athletes
	Facility requirements	Cost of facilities required
	Historical Legacy	Event count from previous Olympics
Gender Equity	Gender event ratio	Ratio of male to female events
	Athlete gender ratio	Ratio of male to female athletes
Sustainability	Carbon emission	Level of carbon emission
	Green certification	Official record of low environmental impact
	Resource recycling	Level of wasted materials
Inclusivity	Participation diversity	Number of countries involved
	Participation of disadvantaged groups	Inclusion in Paralympics
	Medalist diversity	Number of countries at the podium
Relevance and Innovation	Technological innovation	Use of latest technologies
	Ratio of young audience	Percentage of youth audience
	E-sport incorporation	Inclusion in Olympic Esports Series
Safety and Fair Play	Anti-doping appliance	Number of stimulant cases
	Judging Fairness	Advanced judging system
	Safety risk	Risk of athlete injury

4.2 TOPSIS Analysis based on AHP-EWM Model

Our evaluation model integrates criteria weights resulted from **Analytic Hierarchy Process (AHP)** and **Entropy Weight Model (EWM)**, respectively, with **Grey Relational Analysis (GRA)** and use **Technique for Order Preference by Similarity to Ideal Solution (TOPSIS)** to rank SDEs based on the combined weights.

The **AHP** method structures the problem into a hierarchy of goal, criteria, and subcriteria, subjectively assigning relative importance to each factor through pairwise comparisons^[2]. Consistency checks ensure the judgments are logical, and the eigenvector of the comparison matrix provides the weights for the criteria.

The **EWM** model refines the subcriteria weights by analyzing the variability of data; attributes with higher variability are deemed more informative and assigned greater importance^[3].

Then, the **GRA** evaluates the relative distance of results from the AHP and EWM from the ideal weight and

Finally, **TOPSIS** ranks the sports by calculating their proximity to an ideal solution (best performance across all criteria) and an anti-ideal solution (worst performance)^[4]. By combining these methods, the model balances expert opinions, objective data patterns,

and a comprehensive ranking approach to deliver consistent results.

The overall process is demonstrated in Figure 1.

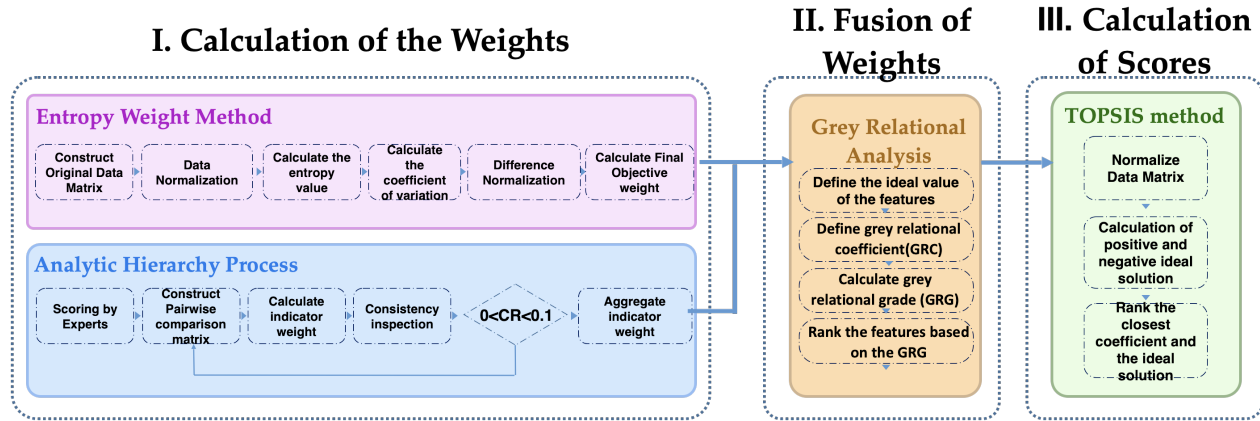


Figure 1: AHP+EWM - TOPSIS Flowchart

4.2.1 Analytical Hierarchy Process for Weight Calculation

1) Constructing the judgment matrix through pairwise comparisons

Each of the criteria is compared in pairs, and the decision-maker provides a subjective judgment on their relative importance. Let there be p evaluation criteria, denoted as:

$$I_1, I_2, \dots, I_p$$

In the following matrix, a_{ij} represents the relative importance comparison between the i -th evaluation criteria and the j -th evaluation criteria (where $i, j = 1, 2, \dots, m$, and m is the number of evaluation criteria).

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mm} \end{bmatrix}$$

Additionally, Table 3 shows the scale of criteria we use to create the matrix.

We review experts' opinions and assign a value for each pair comparison. For example, the relative importance of *Sustainability* compared to *Popularity & Accessibility* might be rated as 3, meaning that *Sustainability* is moderately more important than *Popularity & Accessibility*. The matrix is symmetric, with all diagonal elements equal to 1, indicating equal importance when compared to themselves. Table 4 shows the results of all pairwise comparison matrix of criteria.

Table 3: The Fundamental Scale for Pairwise Comparisons

Intensity of Importance	Definition	Explanation
1	Equal Importance	Two elements contribute equally to the objective
3	Moderate Importance	Experience and judgment moderately favor one element over another
5	Strong Importance	Experience and judgment strongly favor one element over another
7	Very Strong Importance	One element is favored very strongly over another, its dominance is demonstrated in practice
9	Extreme Importance	The evidence favoring one element over another is of the highest possible order of affirmation
Intensities of 2, 4, 6, and 8 can be used to express intermediate values. Intensities of 1.1, 1.2, 1.3, etc. can be used for elements that are very close in importance.		

Table 4: Pairwise Comparison Matrix of Criteria

Criteria	Popularity & Accessibility	Gender Equality	Sustainability	Relevance & Innovation	Safety & Fairness	Inclusivity
Popularity & Accessibility	1	2	1/3	3	4	2
Gender Equality	1/2	1	1/5	2	3	2
Sustainability	3	5	1	4	4	3
Relevance & Innovation	1/3	1/2	1/4	1	2	2
Safety & Fairness	1/4	1/3	1/4	1/2	1	1/2
Inclusivity	1/2	1/2	1/3	1/2	2	1

The judgment matrix for the subcriteria is similarly constructed, shown in Table 5 with the indicator *Inclusivity* as an example.

Table 5: Pairwise Comparison Matrix for Inclusivity Subcriteria

Subcriteria	Global Diversity	Participation of Disadvantaged Groups	Medalist Diversity
Global Diversity	1	3	4
Participation of Disadvantaged Groups	1/3	1	2
Medalist Diversity	1/4	1/2	1

2) Performing consistency checks on the judgment matrix

Once the judgment matrix is constructed, a consistency check is necessary to ensure

that the pairwise comparisons are logically consistent. This is done by calculating the **Consistency Index (CI)** and **Consistency Ratio (CR)**, which helps to measure the consistency of the pairwise comparisons. If the consistency ratio is below a threshold (typically 0.1), the judgments are considered consistent enough for reliable analysis.

The consistency index (CI) and consistency ratio (CR) are calculated as follows:

$$CI = \frac{\lambda_{\max} - p}{p - 1}, \quad CR = \frac{CI}{RI}$$

where $\lambda_{\max}$ is the largest eigenvalue of the matrix, p is the number of criteria, and RI is the random consistency index.

The CI is calculated to measure how consistent the judgments are, while the CR provides an overall consistency ratio. The judgment matrix is considered sufficiently consistent for further analysis if the CR is below 0.1.

3) Calculating the weights of each criteria

After ensuring consistency, the next step is to calculate the weights of each criteria. This is done by first calculating the geometric mean of each row in the judgment matrix and then normalizing the values to obtain the relative weights of the criteria. These weights indicate the importance of each criteria in the decision-making process.

$$G_i = \sqrt[p]{a_{i1} \times a_{i2} \times \cdots \times a_{ip}}, \quad \alpha_i = \frac{G_i}{\sum_{j=1}^p G_j}$$

4) AHP Weight Results

The relative weights determined by AHP are shown below, with Level I indicator weights in Table 3 and subcriteria weights under *Inclusivity* as an example for the Level II indicator groups in Table 4.

Table 6: Criteria Weights from AHP

Criteria	Weight
Popularity & Accessibility	0.2081
Gender Equity	0.1368
Sustainability	0.4121
Inclusivity	0.0993
Relevance & Innovation	0.0554
Safety & Fair Play	0.0884

Table 7: Example of Subcriteria Weights from AHP

Factor	Weight
Global Diversity	0.6046
Participation of Disadvantaged	0.2906
Medalist Diversity	0.1048

4.2.2 Entropy Weight Method for Weight Calculation

The Entropy Weight Method (EWM) is an objective weighting approach commonly used to minimize biases from subjective factors. It calculates indicator weights based on their variability. Lower entropy implies higher variability, less chaotic information, and a higher weight in the overall evaluation. The EWM is implemented through the following steps.

1) Data Normalization

To make indicators comparable, raw data is normalized. Given k indicators $x_1, x_2, \dots, x_k$, with corresponding data sets $X_i = \{x_1, x_2, \dots, x_n\}$, the normalized value Y_{ij} is calculated as:

$$Y_{ij} = \frac{X_{ij} - \min(X_i)}{\max(X_i) - \min(X_i)}.$$

The values for each feature after normalization are shown in Table 8.

Table 8: Normalized Values for EWM Analysis (Simplified)

Sport	Audience Interest	Athlete Numbers	Carbon Emissions	Safety Risk
BOX	0.12	0.11	0.14	0.06
MTB	0.00	0.00	0.09	0.03
DIV	0.00	0.04	0.18	0.15
FEN	0.01	0.09	0.14	0.14
BDM	0.01	0.06	0.14	0.12
⋮	⋮	⋮	⋮	⋮
SHO	0.03	0.17	0.05	0.16
FBL	0.56	0.27	0.00	0.00

2) Calculating Information Entropy

Using information theory, the entropy for each indicator j is calculated as:

$$E_j = -\frac{1}{\ln n} \sum_{i=1}^n p_{ij} \ln p_{ij},$$

where

$$p_{ij} = \frac{Y_{ij}}{\sum_{i=1}^n Y_{ij}}.$$

If $p_{ij} = 0$, it is conventionally defined that $\lim_{p_{ij} \rightarrow 0} p_{ij} \ln p_{ij} = 0$. Indicators with less variation (e.g., few countries participate or low carbon emissions) have higher entropy values and, thus, lower weights in the final ranking. Conversely, indicators with more variability have lower entropy and higher weights.

3) Determining Indicator Weights

The weight of each indicator i is determined using:

$$\beta_i = \frac{1 - E_i}{k - \sum_{j=1}^k E_j}, \quad i = 1, 2, \dots, k.$$

The results, after normalizing and applying the entropy calculations, provide objective weights for each indicator, which are used to rank the sports based on their performance across these dimensions. Examples of the resulting weights are shown in Table 9.

Table 9: Example Results of Weights from the Entropy Weight Method

Factor	Weight
Athlete Gender Ratio	0.0155
Event Gender Ratio	0.0179
Stimulant Use	0.0182
...	...
Safety Risk	0.0256
Carbon Emissions	0.0278
Number of Countries	0.0344

4.2.3 Grey Relational Analysis for Weight Combination

The AHP is a subjective weighing method, while the EWM is more objective, weighing based on data. To combine the two methods, we use Grey Relational Analysis (GRA) to integrate different perspectives on the relative importance of factors.

1) Calculation of GRA

The GRA quantifies the similarity between the normalized weights from AHP and EWM. It is calculated using the following formula:

$$GRA_i = \frac{\sum_{j=1}^n \left(\left| w'_i - e'_j \right| + \xi \cdot \max \left(\left| w'_i - e'_j \right| \right) \right)}{\sum_{j=1}^n \left(\min \left(\left| w'_i - e'_j \right| \right) + \xi \cdot \max \left(\left| w'_i - e'_j \right| \right) \right)}$$

where w'_i are the normalized AHP weights, e'_j are the normalized EWM weights, ξ is a distinguishing coefficient (set to 0.5), and the absolute difference between w'_i and e'_j is used to measure similarity.

For example, when evaluating the factor *audience interest*, the GRA calculates the degree of similarity between the AHP weight (0.6370) and the EWM weight (0.1558) to determine the combined weight for this factor.

2) Final Composite Weights

After computing the GRA values, the final composite weights are calculated using a weighted average formula:

$$w_{final} = \alpha \times AHP + (1 - \alpha) \times EWM$$

where α is the weight coefficient derived from the GRA:

$$\alpha_i = \frac{GRA_i}{\sum_{i=1}^n GRA_i}$$

The composite weight for each factor is then determined by combining the AHP and EWM weights according to their respective grey relational degree, as shown in Table 10. The results are shown in Figure 2, with *Global Audience Interest* and *Resource Recycling* being the most weighted indicators when evaluating the SDEs.

Table 10: Final Weight For Each Factor

Factor	Score
Global Audience Interest	0.193762
Athlete Participation	0.836318
Facility Requirements	0.052435
Event Gender Ratio	0.058245
Athlete Gender Ratio	0.044208
...	...
E-sport Incorporation	0.016107
Anti-Doping Compliance	0.989609
Judging Fairness	0.081316
Safety Risks	0.272218

The GRA approach allows for a more balanced integration of both methods, ensuring that the final weights reflect the relative importance of the factors based on the similarity of their AHP and EWM-derived values.

4.2.4 TOPSIS Evaluation Model

After establishing the weighted criteria through a GRA integration of the AHP and EWM, we apply the TOPSIS model to rank the SDE alternatives. This model helps identify the most suitable SDEs by determining their proximity to both the ideal and anti-ideal solutions and considering both favorable and unfavorable factors.

Using the normalized data, we need to calculate the Weighted Normalized Decision Matrix $W(t)$, which are multiplied element-wise by the normalized matrix to form the

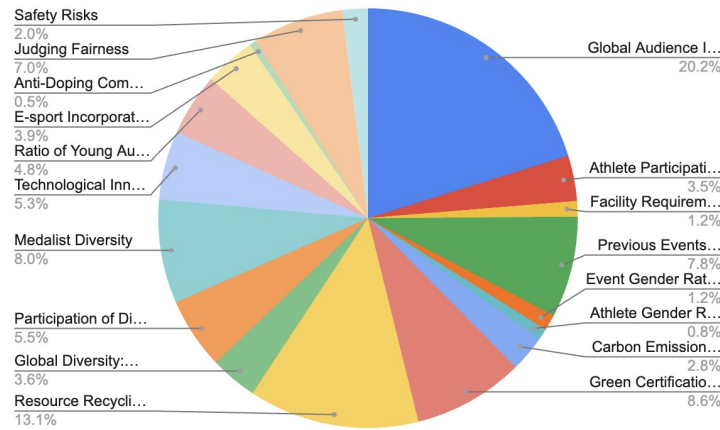


Figure 2: Indicator Contribution Based on Weights from Grey Relational Analysis

weighted normalized decision matrix $W(t)$:

$$W(t) = W_j \cdot Z(t)$$

where W_j is the weight assigned to the j -th criteria, and $Z(t)$ is the normalized decision matrix.

The ideal solution W_j^+ is defined as the maximum value in each column of the weighted normalized matrix, while the anti-ideal solution W_j^- is defined as the minimum value in each column. These solutions represent the best and worst possible outcomes across all criteria.

$$W_j^+ = \begin{cases} \max\{R_{ij}\}, & \text{if } j \text{ is a benefit criteria,} \\ \min\{R_{ij}\}, & \text{if } j \text{ is a cost criteria.} \end{cases}$$

$$W_j^- = \begin{cases} \min\{R_{ij}\}, & \text{if } j \text{ is a benefit criteria,} \\ \max\{R_{ij}\}, & \text{if } j \text{ is a cost criteria.} \end{cases}$$

The distance D_i^+ measures how far the i -th alternative is from the ideal solution, and D_i^- measures the distance from the anti-ideal solution. These distances are calculated using the Euclidean distance formula:

$$D_i^+ = \sqrt{\sum_{j=1}^n (W_{ij} - W_j^+)^2} \quad \text{and} \quad D_i^- = \sqrt{\sum_{j=1}^n (W_{ij} - W_j^-)^2}.$$

The final step involves calculating the relative closeness C_i of each alternative to the ideal solution. This is done by computing the ratio of the distance to the anti-ideal solution

to the total of the distances to the ideal and anti-ideal solutions:

$$C_i = \frac{D_i^-}{D_i^+ + D_i^-}$$

The value of C_i ranges from 0 to 1, where values closer to 1 indicate a better alternative. The SDEs are then ranked based on their C_i scores.

Finally, based on the calculated scores, we rank all the SDEs from the most to the least favorable. The SDE with the highest C_i value is considered the optimal choice.

5 Model Application and Results

In this section, we evaluate the robustness of the proposed model by applying it to a diverse set of SDEs from various Olympic periods, specifically focusing on those added or removed in recent Olympic Games (2020, 2024, and 2028) and those that have been continuously included since 1988 or earlier. We test the model's applicability in assessing a wide range of SDEs, and compare results to their current Olympic status to evaluate the robustness of the model. This will give insight to its potential for guiding future Olympic inclusion and removal decisions.

5.1 Selection of SDEs

We select a diverse group of SDEs, including:

- **Recent SDEs** that are added or removed in the 2020, 2024, or 2028 Olympics include Karate, Breaking, Baseball, Softball, and Skateboarding.
- **Long-standing SDEs** that have been consistently included in the Olympics since 1988 or earlier include Swimming, Table Tennis, and Archery.

This selection allows for a broad examination of the model's ability to evaluate both emerging and established sports, offering insights into how well the model aligns with the Olympic values and decisions about sport inclusion or exclusion.

5.2 Evaluation of Olympic Status

We collect data for selected SDEs, with examples of the raw data presented in Table 11.

Table 11: Examples of Raw Data for SDE Evaluation

Data	BSB	SBL	KTE	SKB
Popularity Accessibility				
Global Audience Interest	15 ^[5]	2 ^[5]	2 ^[5]	2 ^[5]
Athlete Participation	144	144	80	80 ^[6]
Facility Requirement	3 ^[7]	3 ^[8]	1	2 ^[9]
Safety and Fairness				
Judging System	8 ^[10]	8 ^[10]	7 ^[11]	8 ^[12]
Num of Cases of Stimulant Use	2 ^[13]	1 ^[13]	2 ^[13]	0 ^[13]
Safety Risk	0.11	0.13	0.19	0.21 ^[14]
...	...	...	...	...

The raw data is then processed to construct the normalized decision matrix:

$$\mathbf{X}(t) = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mn} \end{pmatrix} = \begin{pmatrix} 0.1193 & 0.1086 & \cdots & 0.0877 \\ 0.0000 & 0.0000 & \cdots & 0.1057 \\ \vdots & \vdots & \ddots & \vdots \\ 0.5596 & 0.2667 & \cdots & 0.1011 \end{pmatrix}$$

5.2.1 Model Results

The ranked scores of tested SDEs are listed in Table 12, and the scores of each SDE's indicators are displayed in Figure 3.

Table 12: TOPSIS Scores of Different Sports

Sport Type	Score
Table Tennis (TTE)	0.5789
Swimming (SWM)	0.5358
Archery (ARC)	0.4986
Baseball (BSB)	0.4838
Karate (KTE)	0.4737
Breaking (BKG)	0.4006
Skateboarding (SKB)	0.3978
Softball (SBL)	0.3145

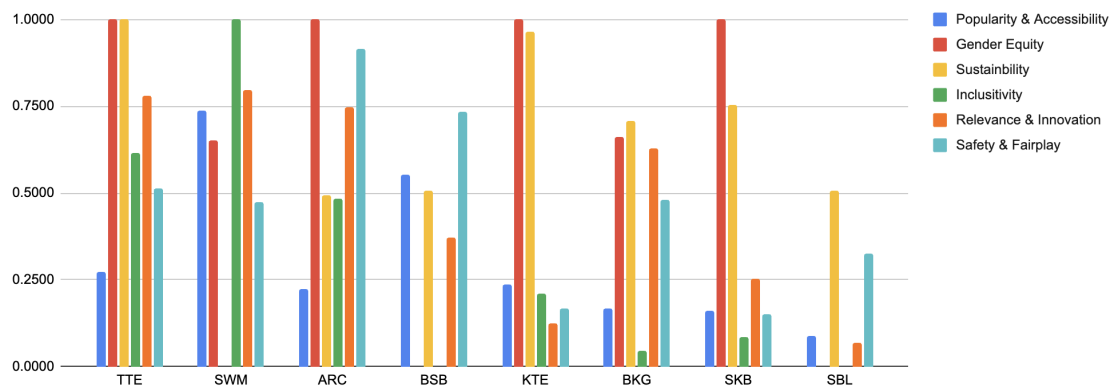


Figure 3: Indicator Scores Presented in Column Chart

5.2.2 Evaluation Analysis

Our model's evaluation result is consistent with the Olympic status of the evaluated sports. It ranks long-standing SDEs in top three and recently added or removed SDEs in a mixed ranking. This reflects the uncertain status of sports with recent changes and confirms the established legacy of long-standing sports in the Olympics. We discuss each sport's ranking interpretation in detail below.

1) Recently Added SDEs

Baseball (BSB) [Rank 4]: Baseball's return after being removed post-2008 highlights its global support, especially in Japan and the U.S. Its reintroduction in 2020 and planned participation in 2028 has boosted its relevance and score.

Skateboarding (SKB) [Rank 7]: Skateboarding, introduced in 2020, appeals strongly to younger audiences and is a sport with significant youth culture representation. Its relatively low ranking reflects its early stage in the Olympics, but continued inclusion in 2024 and 2028 is expected to increase its recognition and popularity over time.

Softball (SBL) [Rank 8]: Softball ranks lower in our model due to its regional popularity, but its return represents cultural significance, especially in countries like the U.S. and Japan, where the sport has deep roots. Its reintroduction strategically appeals to these key markets, and it is expected to grow in global influence, improving its ranking over time.

2) Recently Removed SDEs

Karate (KTE) [Rank 5]: Karate's removal from the Olympics after 2020 reflects its struggle to achieve sustained global appeal compared to other sports. Despite a strong following in certain regions like Japan, its relatively lower score in inclusivity suggests that it failed to garner broad international support. Its removal may signal the IOC's fo-

cus on sports with a more universal presence.

Breaking (BKG) [Rank 6]: Breaking, added for 2024, holds a niche status, especially in youth and urban cultures. However, its lower score and ranking reflect limited traditional Olympic appeal and less global integration. Its removal after the 2024 Olympics may be due to challenges in maintaining widespread and consistent interest, despite its popularity in specific subcultures.

3) Long-standing SDEs

The three long-standing SDEs are the top three of our ranking, with Table Tennis (TTE) having the highest score of 0.5789, Swimming (SWM) being close second, and Archery (ARC) being third. This both confirms the correctness of the model and reflects these sports' widespread popularity and significant cultural impact.

5.3 Sensitivity Analysis

We conduct a sensitivity analysis to assess the robustness of the model by examining how weight shifts of different factors influence the final rankings of SDEs. Specifically, we adjust the weights of each feature by $\pm 5\%$ and $\pm 10\%$, and then analyze the resulting scores and rankings of each SDE. Some examples of resulting scores from weight variation are shown in Table 13.

Table 13: SDE Scores with Weight Adjustments (Partial)

Sport	Stimulant Use +5%	Stimulant Use -5%	Athlete Gender Ratio +5%	Athlete Gender Ratio -5%	...
ARC	0.6568	0.6570	0.6568	0.6570	...
TTE	0.5732	0.5733	0.5732	0.5733	...
SWM	0.5090	0.5090	0.5091	0.5090	...
...	...	...	...	...	...

Individual SDE scores slightly fluctuate ($< 2\%$) with each weight variation, as shown in Figure 4 with Archery (ARC) as an example. Figure 5 shows the line graph of the scores for each SDE, with different line colors revealing different weight variations. The lines fluctuates negligibly and resulted rankings remain the same, suggesting robustness against shifts in the weights of the model.

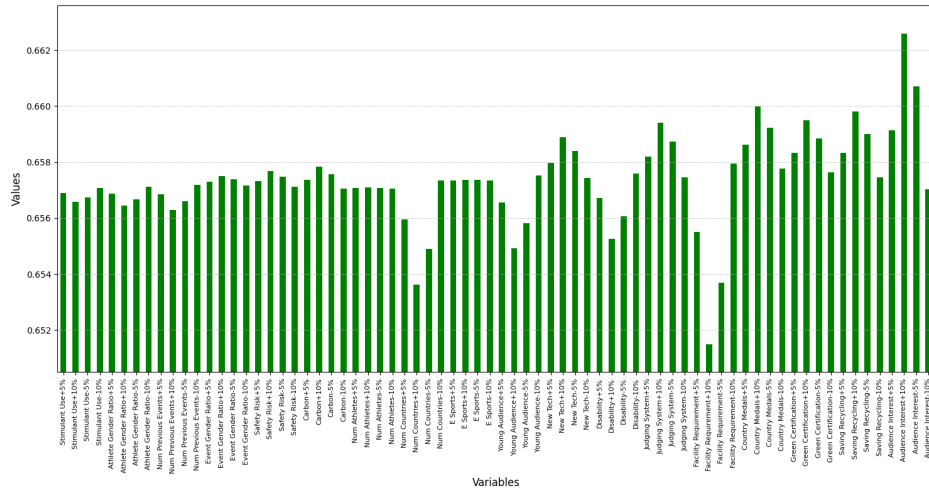


Figure 4: Example of Weight Variation and Corresponding Scores

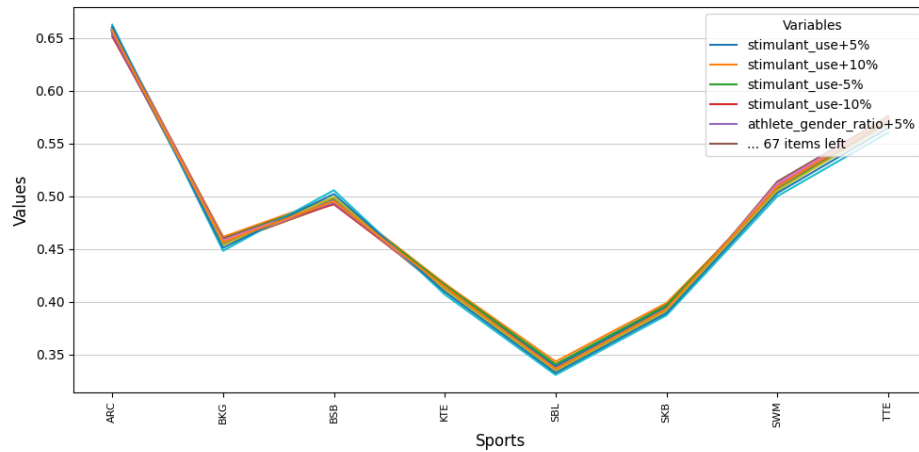


Figure 5: Ranking and Score Variation of Evaluated SDEs

6 Sports Evaluation for Future Olympics

In this section, we consider potential candidates for the **2032 Brisbane Olympics**, including new sports and all Olympic sports absent in the 2028 Games. New SDE candidates are selected based on their cultural connection to the host city, established international competitions, and their current global interest.

To evaluate SDEs with our developed evaluation model for the **2036 Olympics**, the **AutoRegressive Integrated Moving Average (ARIMA)** model^[15] can be used to statistically predict the data of each SDE indicator in 2032 based on historical data. However, due to limited continuous time series data for other indicators, we only use ARIMA to predict **previous event counts** and assume that other indicator values in 2024 can reflect a sport's status in 2032 and 2036 as well. Since there is no recent Olympic data for most of these SDEs, we obtain data for each SDE from their largest competitions worldwide.

6.1 Data Prediction with ARIMA

The ARIMA model predicts future data integrating three key components: **AutoRegressive (AR)**, which uses past values to predict future ones; **Integrated (I)**, which accounts for differences to make the data stationary; and **Moving Average (MA)**, which relies on past errors to enhance predictions. The general form of the ARIMA model is expressed as:

$$X_t = \psi_0 + \sum_{i=1}^p \phi_i X_{t-i} - \sum_{j=1}^q \theta_j \epsilon_{t-j} + \epsilon_t,$$

where X_t is the current value of the time series, ψ_0 is a constant representing the trend component, ϕ_i are the AR coefficients, and θ_j are the MA coefficients. The terms ϵ_t and ϵ_{t-j} represent the random errors at time t and their lagged values, respectively. The parameters p and q define the order of the AR and MA components.

In this study, we use an **ARIMA(1, 1, 0)** model. The AR(1) term incorporates one lagged value of the series X_t , while the I(1) term applies first-order differencing to make the series stationary:

$$Y_t = X_t - X_{t-1}.$$

No moving average component (MA(0)) was included, as no significant residual patterns were identified in the data.

For each sport, represented as rows in the dataset, the ARIMA model was trained using historical event counts starting from the second data point onward to ensure consistency. The prediction for 2032 was generated by forecasting one step ahead:

$$\hat{X}_{t+1} = \psi_0 + \phi_1 X_t + \epsilon_{t+1}.$$

The predicted values were refined by adding 0.4, which accounts for observed patterns in the historical data.

6.2 Evaluation Results

A total of 17 sports are evaluated by our evaluation model, including 7 new sports and all 10 sports that have Olympic legacy but is discontinued in the 2028 Games. The results are shown in Table 14, with **boxing** ranked first, **flying disc** second, and **Australian rules football** third for inclusion in the 2032 Brisbane Olympics.

Assuming we add those sports to the 2032 Olympics, we now consider the rest of the candidates and use the ARIMA model to predict their data in 2032. As noted before, it is most practical for us to only predict the event count factor.

The 14 sports are evaluated with predicted data by our evaluation model, as shown in

Table 14: SDEs Ranking for 2032

Sport	Final Score
Boxing (removed in 2028)	0.6599
Flying Disc (new)	0.5314
Australian Rules Football (new)	0.4970
Bowling (new)	0.4864
Breaking (new)	0.4685
...	...

Table 15. This ranking suggests that **bowling**, **breaking**, and **polo**—ranked first, second, and third, respectively—have the highest potential to be added to the 2036 Olympics.

Table 15: SDEs Ranking for 2036

Sport	Final Score
Bowling (new)	0.6117
Breaking (removed in 2028)	0.4931
Polo (removed in 1936)	0.4902
Tug of War (removed in 1920)	0.4868
Netball (New)	0.4690
...	...

7 Evaluation of Strengths and Weaknesses

7.1 Strengths

Our model uses a multi-faceted approach in building a robust evaluation system. By combining the Analytical Hierarchy Process and Entropy Weight Method, it effectively integrates both subjective expert judgment and objective data in the weights using the Grey Relational Analysis which determines which one of the two is more ideal for each indicator. Finally, the TOPSIS method assesses all SDEs with the established weights and provides a comprehensive ranking that informs decision making for the Olympics.

7.2 Weaknesses

There might be inaccuracies in our data samples of subjective indicators. It is impractical to represent all experts' opinions in one indicator value and some subjectiveness is ineluctable. Furthermore, our justified assumptions might add inaccuracies to our model. For instance, we assume most indicators remain constant throughout predicted years because of limited access to historical data. This does not take into account possible shifts

in other SDE indicators in the time interval, and might have caused inaccuracies in the rankings.

7.3 Future Improvements

In future development of the model, collecting historical data for each sport indicator can enhance the accuracy of our predictions for the 2036 and future Olympics. A comprehensive dataset input for our ARIMA prediction model will enable a more precise forecasting of sports' potential inclusion in future Olympic Games.

8 Conclusion

This paper presents a comprehensive model to assist the International Olympic Committee (IOC) in evaluating sports for the 2032 Summer Olympics in Brisbane. Our model offers a comprehensive and flexible framework for evaluating sports, disciplines, and events for the Olympic Games, combining Entropy Weight Method, Analytical Hierarchy Process, and TOPSIS method to assess both quantitative and qualitative key factors. The model balances objective and subjective perspectives and has strong adaptability for further application in evaluating diverse SDEs. Sensitivity analyses and evaluations confirm the model's robustness in identifying sports with the highest potential for inclusion in the 2032 Games. In future iterations, incorporating more robust historical data collection methods and advanced predictive techniques can improve the model's reliability in forecasting future Olympics events.

9 A Letter to the IOC

To the International Olympics Committee:

We are pleased to present our evaluation of potential sports, disciplines, and events (SDEs) for inclusion in the 2032 Brisbane Summer Olympics. Using official data and literature, our team has developed an evaluation model that uses a combination of multiple weighing techniques to ensure objectiveness and reliability.

To assess each SDE's alignment with the Olympic criteria, we include popularity, gender equity, sustainability, inclusivity, relevance, and safety. Our model incorporates both quantitative data, such as participation rates and globalization, and qualitative factors like cultural appeal and technological innovation. Our model's robustness is assured through sensitivity analyses and evaluation tests with a diverse set of recent and long-standing Olympic events.

Applying our comprehensive model, we identify three SDEs that align well with the Olympic criteria and recommend their inclusion in the 2032 Games: boxing, flying disk, and Australian rules football. These sports combine global appeal, cultural relevance, and innovation, with boxing showcasing rich Olympic tradition and competitiveness, flying disc promoting inclusivity and mixed-gender participation, and Australian rules football highlighting the host nation's heritage while attracting new audiences. The inclusion of the three sports will reflect the IOC's continuous pursuit of diversity and innovation.

Furthermore, we combine our model with a time series data prediction model to evaluate potential sport inclusion for the 2036 Olympic Games. The resulted ranking of the model suggest the addition of bowling and return of breaking and polo.

We are confident that our findings will provide valuable insights that support the IOC's efforts to ensure the Olympic Games remain both relevant and impactful for years to come.

Thank you for the opportunity to assist in shaping the future of the Olympics.

Sincerely,
HiMCM Team 15277

References

- [1] I. O. Committee, Factsheet: The Programme of the Games of the Olympiad, Accessed: 2024-11-18, **2024**, <https://stillmed.olympics.com/media/Documents/Olympic-Games/Factsheets/The-programme-of-the-Games-of-the-Olympiad.pdf>.
- [2] Y. Liu et al., *Expert Systems with Applications* **2020**, 161, 113738.
- [3] C. Team, *Corporate Finance Institute* **2024**.
- [4] A. Tewatia, *GeeksforGeeks* **2021**.
- [5] Google Trends, Olympic Trends - Google Trends, Accessed: 2024-11-18, <https://trends.google.com/trends/explore?date=today%203-m&q=olympic&hl=en>.
- [6] Topend Sports, Athletes at the Olympic Games, Accessed: 2024-11-18, **n.d.** <https://www.topendsports.com/athletes/index.htm>.
- [7] Baseball Competition Venue Guidelines, Accessed: 2024-11-18, **n.d.** <https://static.wbssc.org/assets/cms/documents/94ab4fc9-2a82-80b8-589c-a7e05d19edb6.pdf>.
- [8] World Baseball Softball Confederation, Venue and Facilities Manual 2018, Accessed: 2024-11-18, **2018**, <https://static.wbssc.org/wp-content/uploads/Venue-and-Facilities-Manual-2018-ENG.pdf>.
- [9] G. D. Cecco, Worldskate - Skateboarding & Roller Sports - World Skate Facilities, Accessed: 2024-11-19, **2024**, <https://www.worldskate.org/about/facilities.html>.
- [10] Olympics, Baseball Softball: Olympic history, rules, latest updates and upcoming events for the Olympic sport, Accessed: 2024-11-19, **2024**, <https://olympics.com/en/sports/baseball-softball/>.
- [11] N. Olympics, Karate 101: Rules, Accessed: 2024-11-19, **2021**, <https://www.nbcolympics.com/news/karate-101-rules>.
- [12] Olympics.com, How the scoring for skateboarding street will work at Paris 2024, Accessed: 2024-11-19, **2022**, <https://olympics.com/en/news/how-the-scoring-for-skateboarding-street-will-work-at-paris-2024>.
- [13] World Anti-Doping Agency, *World Anti-Doping Agency* **2023**, Accessed: 2024-11-18.
- [14] G. G. Team, The 2024 Paris Olympics: Guide to Security Considerations for Visitors, Accessed: 2024-11-19, **2024**, <https://www.globalguardian.com/global-digest/paris-summer-olympics-security-guide>.
- [15] V. Kotu, B. Deshpande, *Data Science* **2019**.